

Mahjabin Rahman Oishe

Master's Student in Computer Science | Colorado State University

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RESEARCH INTERESTS

My research interests broadly encompass Machine Learning, Deep Learning, and Computer Vision, with a particular emphasis on medical image analysis and intelligent diagnostic systems. I am especially interested in developing reliable, interpretable, and computationally efficient models that can support clinical decision-making and improve healthcare outcomes. Beyond healthcare applications, I am also interested in Human-Computer Interaction and emerging areas of Artificial Intelligence. My long-term goal is to contribute to applied computer science research by designing technically rigorous and practically meaningful solutions for real-world problems.

EDUCATION

Master of Science (M.S.) in Computer Science

Summer 2026 – Present

Colorado State University, Fort Collins, CO, USA

GPA: 4.00 / 4.00

Advisor: **Dr. Bruce Draper**

Bachelor of Science (B.Sc.) in Computer Science & Engineering

Jan 2018 – Sep 2023

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

CGPA: 3.66 / 4.00

Class Rank: **10th out of 180** in the batch

Higher Secondary School Certificate (H.S.C.)

July 2015 – July 2017

Viqarunnisa Noon College, Dhaka, Bangladesh

GPA: 5.00 / 5.00 • Government Scholarship Awardee

Secondary School Certificate (S.S.C.)

Jan 2014 – May 2015

Viqarunnisa Noon School, Dhaka, Bangladesh

GPA: 5.00 / 5.00 • Government Scholarship Awardee

RESEARCH EXPERIENCE

Graduate Research Assistant — Colorado State University

Summer 2026

- Conducting research in the Department of Computer Science under faculty supervision.
- Research areas include Machine Learning, Deep Learning, Computer Vision, and AI applications in medical image analysis.

Undergraduate Thesis Research — RUET

2018 – 2023

- Developed a ViT-CNN fusion architecture for enhanced glaucoma prediction in OCT images, published in Springer Nature 2025.
- Investigated HOG-feature-based ML methods for glaucoma prediction; accepted at ICIEV & IVPR 2023 (London, UK).
- Collaborated with multi-disciplinary teams on 10+ peer-reviewed publications spanning medical imaging, agricultural AI, and NLP.

TEACHING EXPERIENCE

Lecturer — Green University of Bangladesh (GUB)

Dec 2023 – Mar 2025

Department of Computer Science & Engineering, Narayanganj, Dhaka, Bangladesh

- Delivered lectures & Conducted labs for: Algorithm, Data Communication, Machine Learning, Microprocessors & Microcontrollers, Web Programming, Structured Programming.
- Acknowledged as a proactive educator in the Center of Excellence in Teaching and Learning (CETL) program.

PUBLICATIONS

[1] **M. R. Oishe**, S. M. M. Hasan, and M. F. Zibran, “Breaking the mold: ViT-CNN fusion for enhanced glaucoma prediction in OCT images,” in *Artificial Intelligence and Soft Computing*, Springer Nature Switzerland, 2025, pp. 326–344. ISBN: 978-3-031-81596-6.

[2] **M. R. Oishe**, S. M. M. Hasan, A. Y. Srizon, A. H. Efat, N. Jannat, M. Mitu, and M. F. Faruk, “Enhancing glaucoma prediction performance in OCT images using ML based on HOG features,” in *ICIEV & IVPR 2023*, London, UK. (Accepted)

[3] A. H. Efat, S. M. M. Hasan, N. Jannat, **M. R. Oishe**, et al., “Preventing skin cancer through improved skin lesion recognition: An attention-triplet and multi-layer ensemble based CNN approach,” in *BIM 2023*, Dhaka. (Accepted)

[4] M. M. R. Khan, R. B. Arif, M. A. B. Siddique, and **M. R. Oishe**, “Study and observation of the variation of accuracies of KNN, SVM, LMNN, ENN algorithms on eleven datasets,” in *iCEEiCT 2018*, pp. 124–129. DOI: [10.1109/CEEICT.2018.8628041](https://doi.org/10.1109/CEEICT.2018.8628041)

[5] M. M. R. Khan, M. A. B. Siddique, R. B. Arif, and **M. R. Oishe**, “ADBSCAN: Adaptive density-based spatial clustering of applications with noise,” in *iCEEiCT 2018*, pp. 107–111. DOI: [10.1109/CEEICT.2018.8628138](https://doi.org/10.1109/CEEICT.2018.8628138)

[6] R. B. Arif, M. A. B. Siddique, M. M. R. Khan, and **M. R. Oishe**, “Study of variations of accuracies for handwritten digits recognition using CNN,” in *iCEEiCT 2018*, pp. 112–117. DOI: [10.1109/CEEICT.2018.8628078](https://doi.org/10.1109/CEEICT.2018.8628078)

[7] N. Jannat, S. M. M. Hasan, A. Y. Srizon, **M. Oishe**, et al., “Efficient detection of crop leaf diseases: A lightweight CNN approach for enhanced agricultural productivity,” in *ICICT4SD 2023*, pp. 99–103. DOI: [10.1109/ICICT4SD59951.2023.10303536](https://doi.org/10.1109/ICICT4SD59951.2023.10303536)

[8] N. Haque, A. H. Efat, S. M. M. Hasan, N. Jannat, **M. Oishe**, and M. Mitu, “Revolutionizing pest detection: A transfer learning fusion network with attention-triplet and multi-layer ensemble,” in *ICCIT 2023*, pp. 1–6. DOI: [10.1109/ICCIT60459.2023.10441171](https://doi.org/10.1109/ICCIT60459.2023.10441171)

- [9] S. Shafin, A. H. Efat, S. M. M. Hasan, N. Jannat, **M. Oishe**, et al., “Skin lesion classification through sequential triple attention DenseNet,” in *ICCIT 2023*, pp. 1–6. DOI: [10.1109/ICCIT60459.2023.10441280](https://doi.org/10.1109/ICCIT60459.2023.10441280)
- [10] T. S. Joy, A. H. Efat, S. M. M. Hasan, N. Jannat, **M. Oishe**, et al., “Attention Trinity Net and DenseNet fusion for American Sign Language recognition,” in *ICCIT 2023*, pp. 1–6. DOI: [10.1109/ICCIT60459.2023.10440975](https://doi.org/10.1109/ICCIT60459.2023.10440975)
- [11] R. Haque, S. M. M. Hasan, A. Y. Srizon, A. H. Efat, **M. Oishe**, et al., “Evaluating the efficacy of feature selection methods in cardiovascular disease prediction with ML,” in *EICT 2023*, pp. 1–6. DOI: [10.1109/EICT61409.2023.10427833](https://doi.org/10.1109/EICT61409.2023.10427833)
- [12] M. Mitu, S. M. M. Hasan, A. H. Efat, N. Jannat, and **M. Oishe**, “An explainable ML framework for multiple medical datasets classification,” in *NCIM 2023*, pp. 1–6. DOI: [10.1109/NCIM59001.2023.10212821](https://doi.org/10.1109/NCIM59001.2023.10212821)
- [13] S. Sikder, A. H. Efat, S. M. M. Hasan, N. Jannat, **M. Oishe**, et al., “A triple-level ensemble-based brain tumor classification using Dense-ResNet with three attention mechanisms,” in *ICCIT 2023*, pp. 1–6. DOI: [10.1109/ICCIT60459.2023.10441290](https://doi.org/10.1109/ICCIT60459.2023.10441290)

TECHNICAL SKILLS

Languages: Python, C, C++, Java, PHP

ML / AI: Machine Learning, Deep Learning, Transformers, Computer Vision, NLP

Frameworks: TensorFlow, PyTorch, Scikit-learn

Web Dev: HTML, CSS, JavaScript, Apache Web Server

Databases: MySQL, SQLite

Other: L^AT_EX, Git, Jupyter Notebooks

Problem Solving: 300+ problems solved on Codeforces and UVA Online Judge

ONLINE COURSES

- Programming for Everybody: Getting Started with Python — Coursera
- Machine Learning — Coursera
- Neural Networks and Deep Learning — Coursera
- Transformers in Computer Vision — Udemy

AWARDS & CERTIFICATION

- Graduate Research Assistantship — Colorado State University *Summer 2026*
- Mentorship Training Certificate — Colorado State University *2025*
- CITI Program Certification: Research Security Training (Combined) — CSU *2025*
- CITI Program Certification: Human Research (Group 2, Social/Behavioral) — CSU *2025*
- Award of Leadership — Served as Class Representative in College
- Government Scholarship — Higher Secondary Certificate Examination *2017*
- Government Scholarship — Secondary School Certificate Examination *2015*

- Government Scholarship — Junior School Certificate Examination *2012*
- Government Scholarship — Primary School Certificate Examination *2009*

REFERENCES

Dr. Bruce Draper

Professor & Department Chair
Department of Computer Science
Colorado State University
Fort Collins, CO 80523
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Dr. Nathaniel Blanchard

Assistant Professor
Department of Computer Science
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